

## 1. General Description

The EM74HC597; EM74HCT597 is an 8-bit shift register with input flip-flops. It consists of an 8-bit storage register feeding a parallel-in, serial-out 8-bit shift register. Both the storage register and the shift register have positive edge-triggered clocks. The shift register also has direct load (from storage) and clear inputs. Inputs include clamp diodes, this enables the use of current limiting resistors to interface inputs to voltages in excess of  $V_{CC}$ .

## 2. Features and Benefits

- Wide supply voltage range from 2.8 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- 8-bit parallel storage register inputs
- Shift register has direct overriding load and clear
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standard:
  - JESD8C(2.8 V to 3.6 V)
  - JESD7A(2.8 V to 6.0 V)
- Input levels:
  - For EM74HC597: CMOS level
  - For EM74HCT597: TTL level
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

## EM74HC597; EM74HCT597

8-bit shift register with input flip-flops

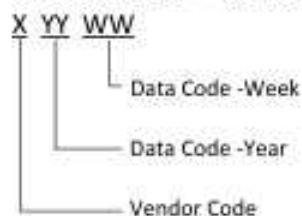
### 3. Ordering Information

Table 1. Ordering information

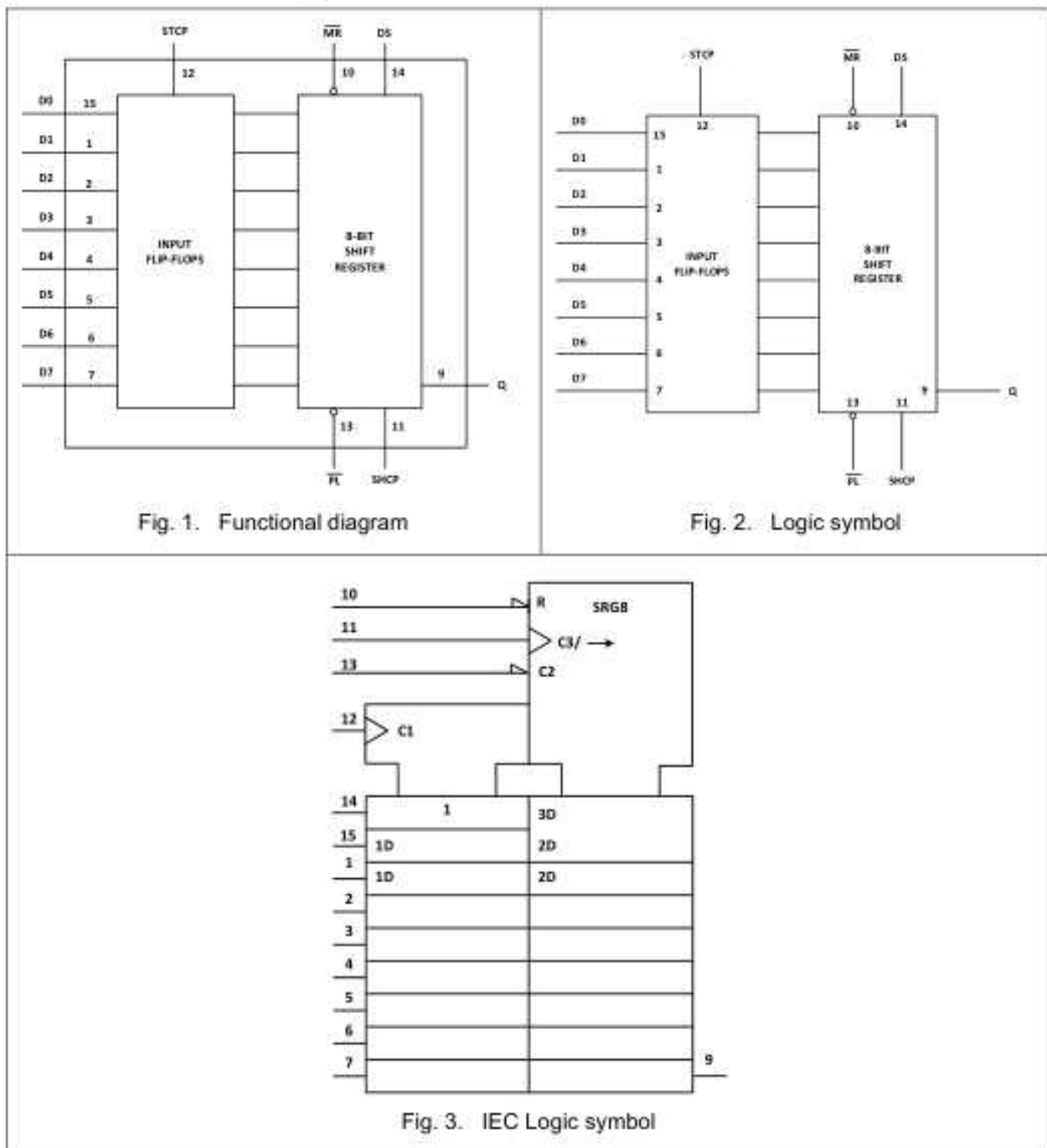
| Type number  | Topside marking | Package   |                                                                           |          |
|--------------|-----------------|-----------|---------------------------------------------------------------------------|----------|
|              |                 | Name      | Description                                                               | Quantity |
| EM74HC597D   | HC597<br>YYYWW  | SOP-16L   | plastic small outline package; 16 leads;<br>body width 3.9 mm             | 3000     |
| EM74HCT597D  | HCT597<br>YYYWW |           |                                                                           |          |
| EM74HC597PW  | HC597<br>YYYWW  | TSSOP-16L | plastic thin shrink small outline package;<br>16 leads; body width 4.4 mm | 3000     |
| EM74HCT597PW | HCT597<br>YYYWW |           |                                                                           |          |

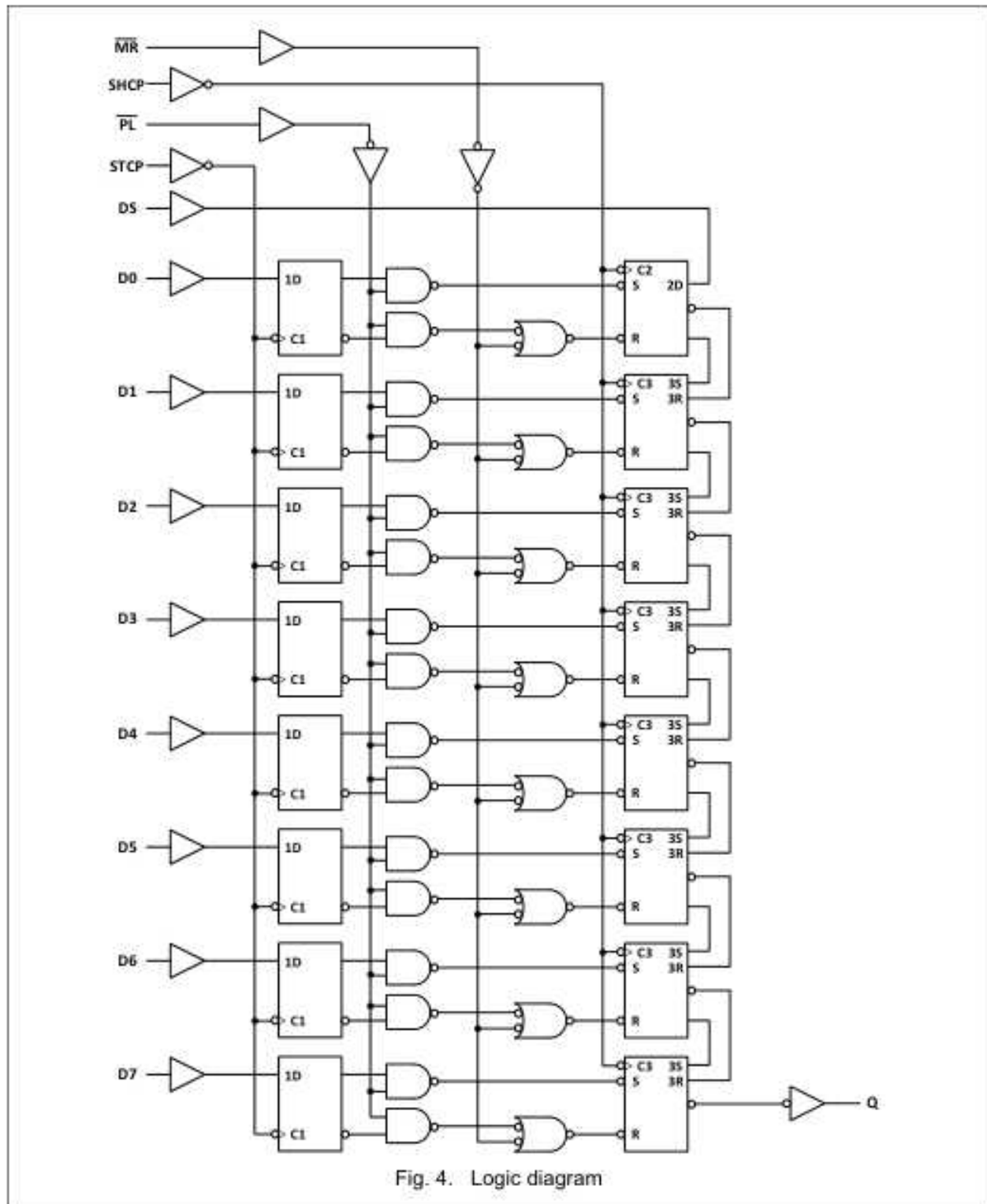
#### MARKING INFORMATION

NOTE: YYYWW = Vendor Code and Data Code.



## 4. Function Diagram





## 5. Pinning Information

### 5.1. Pinning

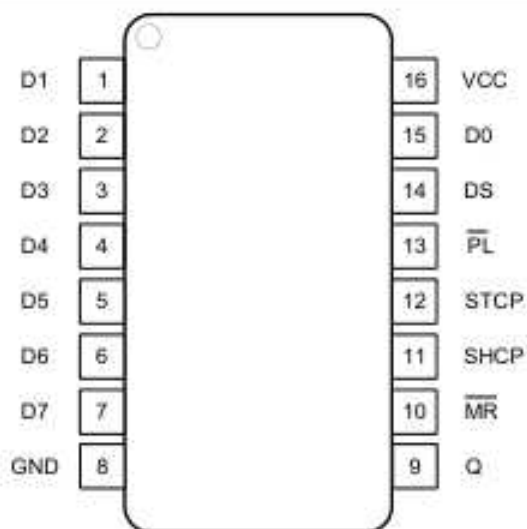


Fig. 5. Top view pin configuration SOP and TSSOP

### 5.2. Pin description

Table 2. Pin description

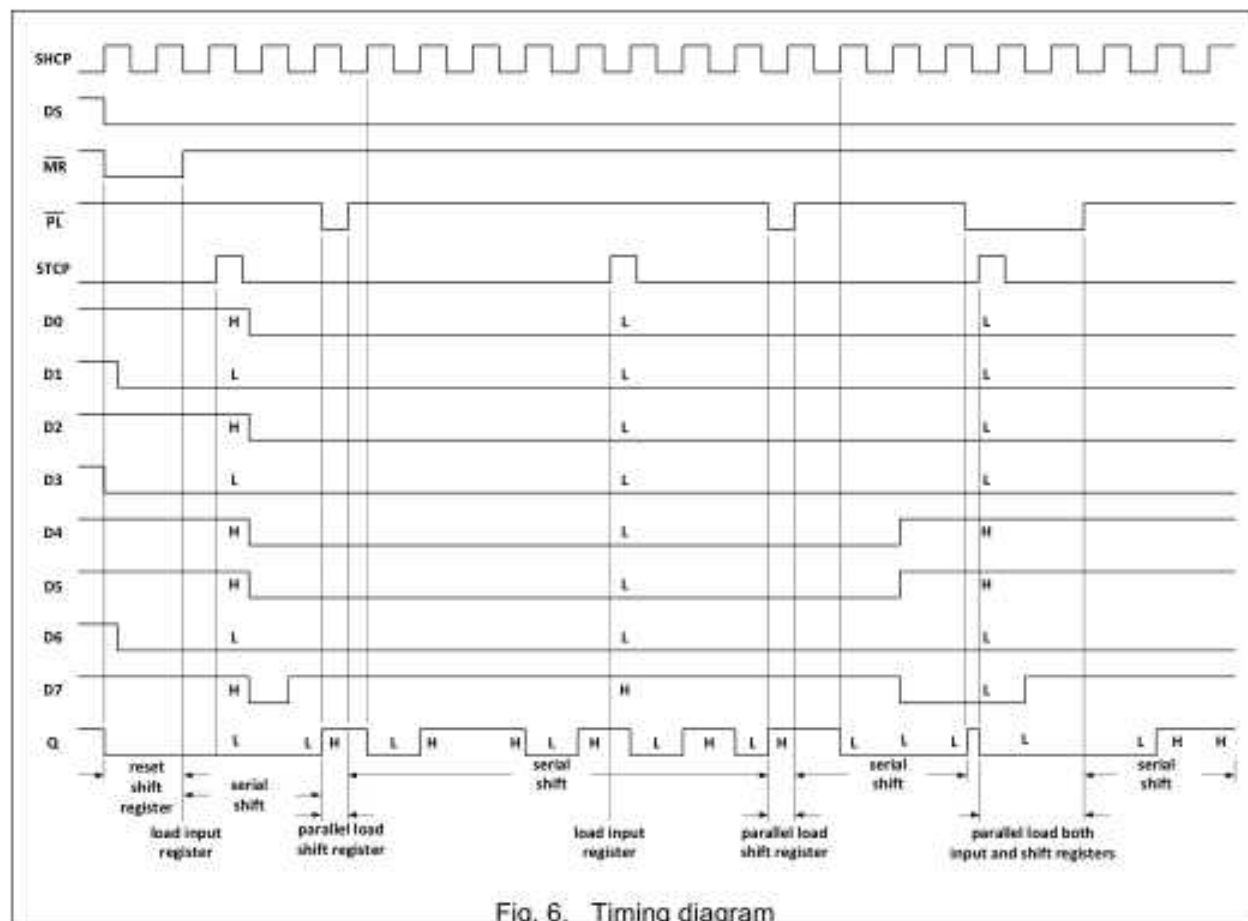
| Symbol                  | Pin              | Description                                                |
|-------------------------|------------------|------------------------------------------------------------|
| GND                     | 8                | ground(0V)                                                 |
| Q                       | 9                | serial data output                                         |
| $\overline{\text{MR}}$  | 10               | asynchronous master reset input (active LOW)               |
| SHCP                    | 11               | shift register clock input (LOW-to-HIGH, edge-triggered)   |
| STCP                    | 12               | storage register clock input (LOW-to-HIGH, edge-triggered) |
| $\overline{\text{PL}}$  | 13               | parallel load input (active LOW)                           |
| DS                      | 14               | serial data input                                          |
| D0,D1,D2,D3,D4,D5,D6,D7 | 15,1,2,3,4,5,6,7 | parallel data inputs                                       |
| V <sub>CC</sub>         | 16               | supply voltage                                             |

## 6. Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level; X = don't care; ↑ = positive-going transition.

| Inputs        |      |                 |                 | Function                                                                     |
|---------------|------|-----------------|-----------------|------------------------------------------------------------------------------|
| STCP          | SHCP | $\overline{PL}$ | $\overline{MR}$ |                                                                              |
| ↑             | X    | X               | X               | data loaded to input latches                                                 |
| ↑             | X    | L               | H               | data loaded from inputs to shift register                                    |
| no clock edge | X    | L               | H               | data transferred from input flip-flops to shift register                     |
| X             | X    | L               | L               | invalid logic, state of shift register is indeterminate when signals removed |
| X             | X    | H               | L               | shift register cleared                                                       |
| X             | ↑    | H               | H               | shift register clocked $Q_n = Q_{n-1}$ , $Q_0 = DS$                          |



## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                                                              | Min  | Max      | Unit             |
|-----------|-------------------------|-------------------------------------------------------------------------|------|----------|------------------|
| $V_{CC}$  | supply voltage          |                                                                         | -0.5 | 7.0      | V                |
| $I_{IK}$  | input clamping current  | $V_I < -0.5 \text{ V}$ or $V_I > V_{CC} + 0.5 \text{ V}$ [1]            |      | $\pm 20$ | mA               |
| $I_{OK}$  | output clamping current | $V_O < -0.5 \text{ V}$ or $V_O > V_{CC} + 0.5 \text{ V}$ [1]            |      | $\pm 20$ | mA               |
| $I_O$     | output current          | $-0.5 \text{ V} < V_O < V_{CC} + 0.5 \text{ V}$                         |      | $\pm 25$ | mA               |
| $I_{CC}$  | supply current          |                                                                         |      | 50       | mA               |
| $I_{GND}$ | ground current          |                                                                         | -50  |          | mA               |
| $P_{tot}$ | total power dissipation | $T_{amb} = -40 \text{ }^\circ\text{C}$ to $+125 \text{ }^\circ\text{C}$ |      | 500      | mW               |
| $T_{stg}$ | storage temperature     |                                                                         | -65  | 150      | $^\circ\text{C}$ |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions               | EM74HC597 |      |          | EM74HCT597 |      |          | Unit             |
|---------------------|-------------------------------------|--------------------------|-----------|------|----------|------------|------|----------|------------------|
|                     |                                     |                          | Min       | Typ  | Max      | Min        | Typ  | Max      |                  |
| $V_{CC}$            | supply voltage                      |                          | 2.8       | 5.0  | 6.0      | 4.5        | 5.0  | 5.5      | V                |
| $V_I$               | input voltage                       |                          | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                |
| $V_O$               | output voltage                      |                          | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                |
| $T_{amb}$           | ambient temperature                 |                          | -40       | +25  | +125     | -40        | +25  | +125     | $^\circ\text{C}$ |
| $\Delta t/\Delta V$ | input transition rise and fall rate | $V_{CC} = 3.0 \text{ V}$ |           |      | 371      |            |      |          | ns/V             |
|                     |                                     | $V_{CC} = 4.5 \text{ V}$ |           | 1.67 | 139      |            | 1.67 | 139      | ns/V             |
|                     |                                     | $V_{CC} = 6.0 \text{ V}$ |           |      | 83       |            |      |          | ns/V             |

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8-bit shift register with input flip-flops

## 9. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

| Symbol          | Parameter                 | Conditions                                                                                | -40 °C to +85 °C |        |      | -40 °C to +125 °C |      | Unit |
|-----------------|---------------------------|-------------------------------------------------------------------------------------------|------------------|--------|------|-------------------|------|------|
|                 |                           |                                                                                           | Min              | Typ[1] | Max  | Min               | Max  |      |
| EM74HC597       |                           |                                                                                           |                  |        |      |                   |      |      |
| V <sub>IH</sub> | HIGH-level input voltage  | V <sub>CC</sub> = 3.0 V                                                                   | 2.0              | 1.8    |      | 2.0               |      | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                   | 3.15             | 2.4    |      | 3.15              |      | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                   | 4.2              | 3.5    |      | 4.2               |      | V    |
| V <sub>IL</sub> | LOW-level input voltage   | V <sub>CC</sub> = 3.0 V                                                                   |                  | 1.3    | 0.6  |                   | 0.6  | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                   |                  | 2.1    | 1.35 |                   | 1.35 | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                   |                  | 2.8    | 1.8  |                   | 1.8  | V    |
| V <sub>OH</sub> | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                       |                  |        |      |                   |      |      |
|                 |                           | I <sub>O</sub> = -20μA; V <sub>CC</sub> = 3.0 V                                           | 2.9              | 3.0    |      | 2.9               |      | V    |
|                 |                           | I <sub>O</sub> = -20μA; V <sub>CC</sub> = 4.5 V                                           | 4.4              | 4.5    |      | 4.4               |      | V    |
|                 |                           | I <sub>O</sub> = -20μA; V <sub>CC</sub> = 6.0 V                                           | 5.9              | 6.0    |      | 5.9               |      | V    |
|                 |                           | I <sub>O</sub> = -4.0 mA; V <sub>CC</sub> = 4.5 V                                         | 3.84             | 4.4    |      | 3.7               |      | V    |
|                 |                           | I <sub>O</sub> = -5.2 mA; V <sub>CC</sub> = 6.0 V                                         | 5.34             | 5.9    |      | 5.2               |      | V    |
| V <sub>OL</sub> | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                       |                  |        |      |                   |      |      |
|                 |                           | I <sub>O</sub> = 20μA; V <sub>CC</sub> = 3.0 V                                            |                  | 0      | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20μA; V <sub>CC</sub> = 4.5 V                                            |                  | 0      | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20μA; V <sub>CC</sub> = 6.0 V                                            |                  | 0      | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 4.0 mA; V <sub>CC</sub> = 4.5 V                                          |                  | 0.04   | 0.33 |                   | 0.4  | V    |
|                 |                           | I <sub>O</sub> = 5.2 mA; V <sub>CC</sub> = 6.0 V                                          |                  | 0.05   | 0.33 |                   | 0.4  | V    |
| I <sub>I</sub>  | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 6.0 V                      |                  | 0.1    | ±1   |                   | ±1   | μA   |
| I <sub>CC</sub> | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND; I <sub>O</sub> = 0A ;<br>V <sub>CC</sub> = 6.0 V |                  | 11     | 20   |                   | 40   | μA   |
| C <sub>I</sub>  | input capacitance         | Pin $\overline{\text{PL}}$ , MR                                                           |                  | 4.5    |      |                   |      | pF   |
|                 |                           | Pin DS, STCP,SHCP                                                                         |                  | 8.0    |      |                   |      | pF   |
|                 |                           | Pin D0 to D7                                                                              |                  | 9.5    |      |                   |      | pF   |

## EM74HC597; EM74HCT597

8-bit shift register with input flip-flops

| Symbol           | Parameter                 | Conditions                                                                                                                               | -40 °C to +85 °C |        |      | -40 °C to +125 °C |     | Unit |
|------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------|------|-------------------|-----|------|
|                  |                           |                                                                                                                                          | Min              | Typ[1] | Max  | Min               | Max |      |
| EM74HCT597       |                           |                                                                                                                                          |                  |        |      |                   |     |      |
| V <sub>IH</sub>  | HIGH-level input voltage  | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                         | 2.0              | 1.7    |      | 2.0               |     | V    |
| V <sub>IL</sub>  | LOW-level input voltage   | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                         |                  | 1.4    | 0.8  |                   | 0.8 | V    |
| V <sub>OH</sub>  | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 4.5 V                                                            |                  |        |      |                   |     |      |
|                  |                           | I <sub>O</sub> = -20μA                                                                                                                   | 4.4              | 4.5    |      | 4.4               |     | V    |
|                  |                           | I <sub>O</sub> = -4.0 mA                                                                                                                 | 3.84             | 4.4    |      | 3.7               |     | V    |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ; V <sub>CC</sub> = 4.5 V                                                            |                  |        |      |                   |     |      |
|                  |                           | I <sub>O</sub> = 20μA                                                                                                                    |                  | 0      | 0.1  |                   | 0.1 | V    |
|                  |                           | I <sub>O</sub> = 4.0 mA                                                                                                                  |                  | 0.04   | 0.33 |                   | 0.4 | V    |
| I <sub>I</sub>   | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ; V <sub>CC</sub> = 5.5 V                                                                        |                  | 0.1    | ±1   |                   | ±1  | μA   |
| I <sub>CC</sub>  | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND ; I <sub>O</sub> = 0A ; V <sub>CC</sub> = 5.5 V                                                  |                  | 11     | 20   |                   | 40  | μA   |
| ΔI <sub>CC</sub> | additional supply current | V <sub>I</sub> = V <sub>CC</sub> - 2.1 V; other inputs at V <sub>CC</sub> or GND; V <sub>CC</sub> = 4.5 V to 5.5 V; I <sub>O</sub> = 0 A |                  |        | 450  |                   | 490 | μA   |
| C <sub>i</sub>   | input capacitance         | Pin PL,MR                                                                                                                                |                  | 4.5    |      |                   |     | pF   |
|                  |                           | Pin DS, STCP,SHCP                                                                                                                        |                  | 8.0    |      |                   |     | pF   |
|                  |                           | Pin D0 to D7                                                                                                                             |                  | 9.5    |      |                   |     | pF   |

[1] All typical values are measured at  $T_{amb} = 25^\circ\text{C}$ .

## 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 13.

| Symbol          | Parameter         | Conditions                                  | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|-----------------|-------------------|---------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|                 |                   |                                             | Min              | Typ[1] | Max | Min               | Max |      |
| EM74HC597       |                   |                                             |                  |        |     |                   |     |      |
| t <sub>pd</sub> | propagation delay | SHCP to Q; see Fig. 7 [2]                   |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     |                  |        | 27  |                   | 30  | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     |                  |        | 17  |                   | 20  | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     |                  |        | 13  |                   | 15  | ns   |
|                 |                   | $\overline{\text{MR}}$ to Q; see Fig. 8 [2] |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     |                  |        | 27  |                   | 30  | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     |                  |        | 17  |                   | 20  | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     |                  |        | 13  |                   | 15  | ns   |
|                 |                   | STCP to Q; see Fig. 7 [2]                   |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     |                  |        | 27  |                   | 30  | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     |                  |        | 17  |                   | 20  | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     |                  |        | 13  |                   | 15  | ns   |
|                 |                   | $\overline{\text{PL}}$ to Q; see Fig. 9 [2] |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     |                  |        | 27  |                   | 30  | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     |                  |        | 17  |                   | 20  | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     |                  |        | 13  |                   | 15  | ns   |
| t <sub>t</sub>  | transition time   | Q; see Fig. 9 [3]                           |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     |                  |        | 8   |                   | 10  | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     |                  |        | 7   |                   | 9   | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     |                  |        | 7   |                   | 9   | ns   |
| t <sub>w</sub>  | pulse width       | STCP HIGH or LOW; see Fig. 7                |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     | 25               |        |     | 31                |     | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     | 20               |        |     | 24                |     | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     | 17               |        |     | 20                |     | ns   |
|                 |                   | SHCP HIGH or LOW; see Fig. 7                |                  |        |     |                   |     |      |
|                 |                   | V <sub>CC</sub> = 3.0 V                     | 25               |        |     | 31                |     | ns   |
|                 |                   | V <sub>CC</sub> = 4.5 V                     | 20               |        |     | 24                |     | ns   |
|                 |                   | V <sub>CC</sub> = 6.0 V                     | 17               |        |     | 20                |     | ns   |

**EM74HC597; EM74HCT597**
**8-bit shift register with input flip-flops**

| Symbol    | Parameter     | Conditions                               | -40 °C to +125 °C |        |     | -40 °C to +125 °C |     | Unit |
|-----------|---------------|------------------------------------------|-------------------|--------|-----|-------------------|-----|------|
|           |               |                                          | Min               | Typ[1] | Max | Min               | Max |      |
| $t_w$     | pulse width   | $\overline{MR}$ LOW; see Fig. 8          |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 25                |        |     | 31                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 20                |        |     | 24                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 17                |        |     | 20                |     | ns   |
|           |               | $\overline{PL}$ LOW; see Fig. 9          |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 25                |        |     | 31                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 20                |        |     | 24                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 17                |        |     | 20                |     | ns   |
| $t_{rec}$ | recovery time | $\overline{MR}$ to SHCP; see Fig. 10     |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 20                |        |     | 23                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 15                |        |     | 18                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 13                |        |     | 15                |     | ns   |
| $t_{su}$  | set-up time   | Dn to STCP; see Fig. 11                  |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 20                |        |     | 23                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 15                |        |     | 18                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 13                |        |     | 15                |     | ns   |
|           |               | DS to SHCP; see Fig. 11                  |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 20                |        |     | 23                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 15                |        |     | 18                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 13                |        |     | 15                |     | ns   |
|           |               | $\overline{PL}$ to SHCP; see Fig. 12     |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 20                |        |     | 23                |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 15                |        |     | 18                |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 13                |        |     | 15                |     | ns   |
| $t_h$     | hold time     | Dn to STCP; see Fig. 11                  |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |
|           |               | $\overline{PL}, DS$ to SHCP; see Fig. 11 |                   |        |     |                   |     |      |
|           |               | $V_{CC} = 3.0\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |
|           |               | $V_{CC} = 4.5\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |
|           |               | $V_{CC} = 6.0\text{ V}$                  | 5                 |        |     | 5                 |     | ns   |

# EM74HC597; EM74HCT597

8-bit shift register with input flip-flops

| Symbol            | Parameter                     | Conditions                                                  | -40 °C to +125 °C |        |     | -40 °C to +125 °C |     | Unit |
|-------------------|-------------------------------|-------------------------------------------------------------|-------------------|--------|-----|-------------------|-----|------|
|                   |                               |                                                             | Min               | Typ[1] | Max | Min               | Max |      |
| $f_{max}$         | maximum frequency             | SHCP; see Fig. 7                                            |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 3.0\text{ V}$                                     | 20                |        |     | 16                |     | MHz  |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 24                |        |     | 20                |     | MHz  |
|                   |                               | $V_{CC} = 6.0\text{ V}$                                     | 28                |        |     | 24                |     | MHz  |
| $C_{PD}$          | power dissipation capacitance | $f = 1\text{ MHz}$ ;<br>$V_i = \text{GND to } V_{CC}$ ; [4] |                   | 26     |     |                   |     | pF   |
| <b>EM74HCT597</b> |                               |                                                             |                   |        |     |                   |     |      |
| $t_{pd}$          | propagation delay             | SHCP to $Q_i$ ; see Fig. 7 [2]                              |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     |                   |        | 17  |                   | 20  | ns   |
|                   |                               | $\overline{MR}$ to $Q_i$ ; see Fig. 8 [2]                   |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     |                   |        | 17  |                   | 20  | ns   |
|                   |                               | STCP to $Q_i$ ; see Fig. 7 [2]                              |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     |                   |        | 17  |                   | 20  | ns   |
|                   |                               | $\overline{PL}$ to $Q_i$ ; see Fig. 9 [2]                   |                   |        |     |                   |     |      |
| $t_i$             | transition time               | $Q_i$ ; see Fig. 9 [3]                                      |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     |                   |        | 7   |                   | 9   | ns   |
| $t_w$             | pulse width                   | STCP HIGH or LOW; see Fig. 7                                |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 20                |        |     | 24                |     | ns   |
|                   |                               | SHCP HIGH or LOW; see Fig. 7                                |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 15                |        |     | 18                |     | ns   |
|                   |                               | $\overline{MR}$ LOW; see Fig. 8                             |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 31                |        |     | 38                |     | ns   |
|                   |                               | $\overline{PL}$ LOW; see Fig. 9                             |                   |        |     |                   |     |      |
| $t_{rec}$         | recovery time                 | $\overline{MR}$ to SHCP; see Fig. 10                        |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 15                |        |     | 18                |     | ns   |
| $t_{su}$          | set-up time                   | $D_n$ to STCP; see Fig. 11                                  |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 15                |        |     | 18                |     | ns   |
|                   |                               | $D_S$ to SHCP; see Fig. 11                                  |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 15                |        |     | 18                |     | ns   |
|                   |                               | $\overline{PL}$ to SHCP; see Fig. 12                        |                   |        |     |                   |     |      |
|                   |                               | $V_{CC} = 4.5\text{ V}$                                     | 15                |        |     | 18                |     | ns   |

# EM74HC597; EM74HCT597

## 8-bit shift register with input flip-flops

| Symbol    | Parameter                     | Conditions                                                | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|-----------|-------------------------------|-----------------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|           |                               |                                                           | Min              | Typ[1] | Max | Min               | Max |      |
| $t_H$     | hold time                     | Dn to STCP; see Fig. 11                                   |                  |        |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                   | 5                |        |     | 5                 |     | ns   |
|           |                               | $\overline{PL}, DS$ to SHCP; see Fig.11                   |                  |        |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                   | 5                |        |     | 5                 |     | ns   |
| $f_{max}$ | maximum frequency             | SHCP; see Fig. 7                                          |                  |        |     |                   |     |      |
|           |                               | $V_{CC} = 4.5\text{ V}$                                   | 24               |        |     | 20                |     | MHz  |
| $C_{PD}$  | power dissipation capacitance | $f = 1\text{ MHz};$<br>$V_i = \text{GND to } V_{CC};$ [4] |                  | 28     |     |                   |     | pF   |

[1] Typical values are measured at  $T_{amb} = 25\text{ °C}$ .

[2]  $t_{FHL}$  is the same as  $t_{PLH}$  and  $t_{HOL}$ .

[3]  $t_i$  is the same as  $t_{THL}$  and  $t_{TLH}$ .

[4]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu\text{W}$ ).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$  where:

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

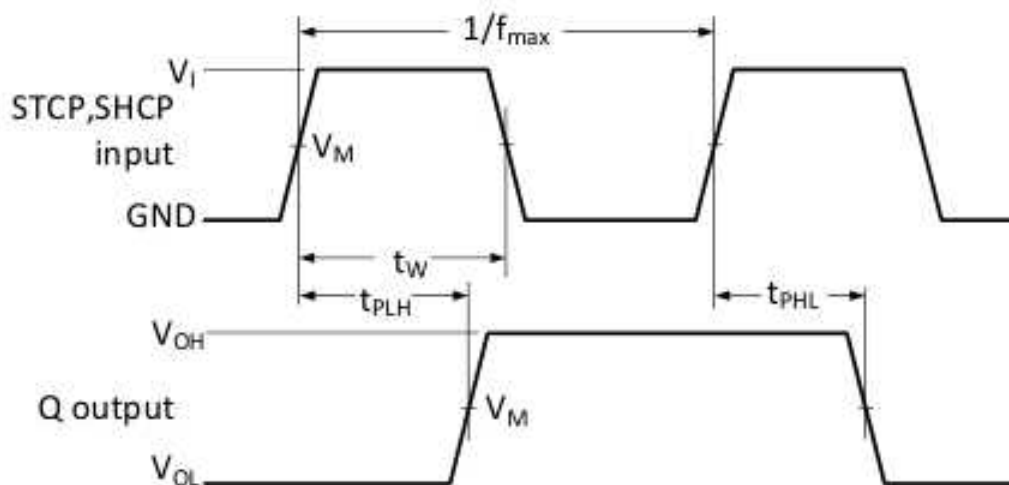
$C_L$  = output load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

$\sum (C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

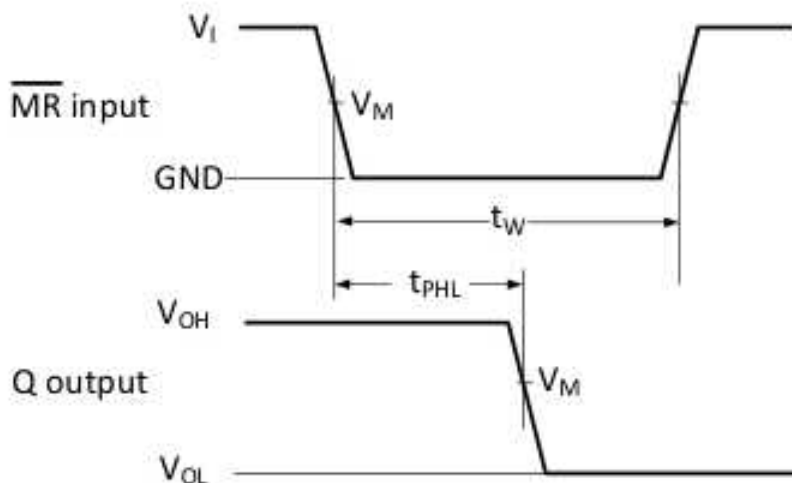
## 11. Waveforms and test circuit



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

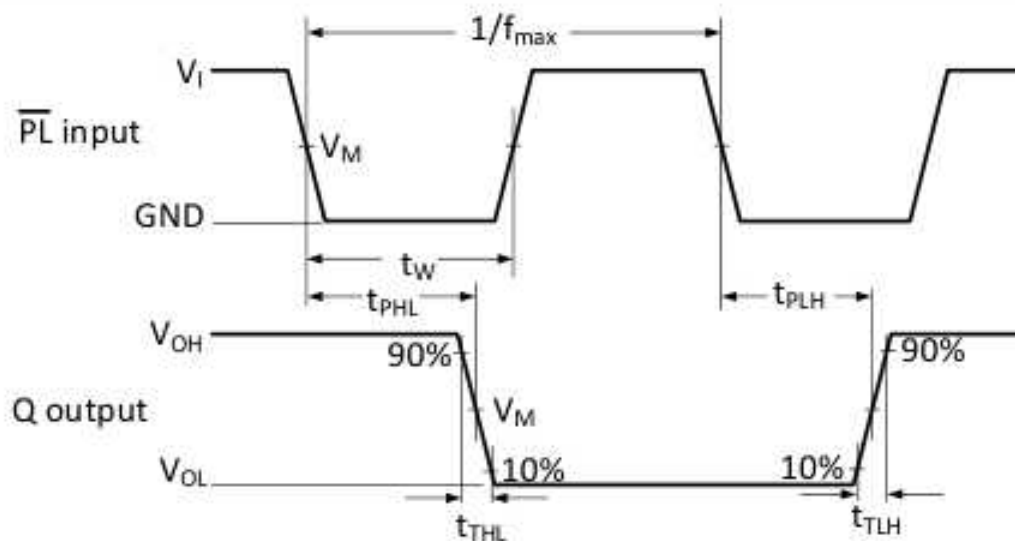
Fig. 7. SHCP and STCP clock inputs to Q output propagation delays, pulse width and maximum clock frequency



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

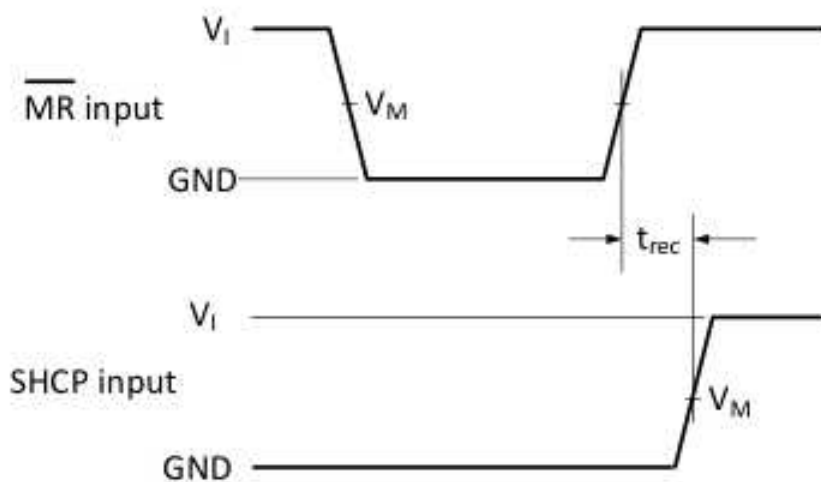
Fig. 8. Input  $\overline{MR}$  to Q output propagation delays and  $\overline{MR}$  pulse width



Measurement points are given in Table 8.

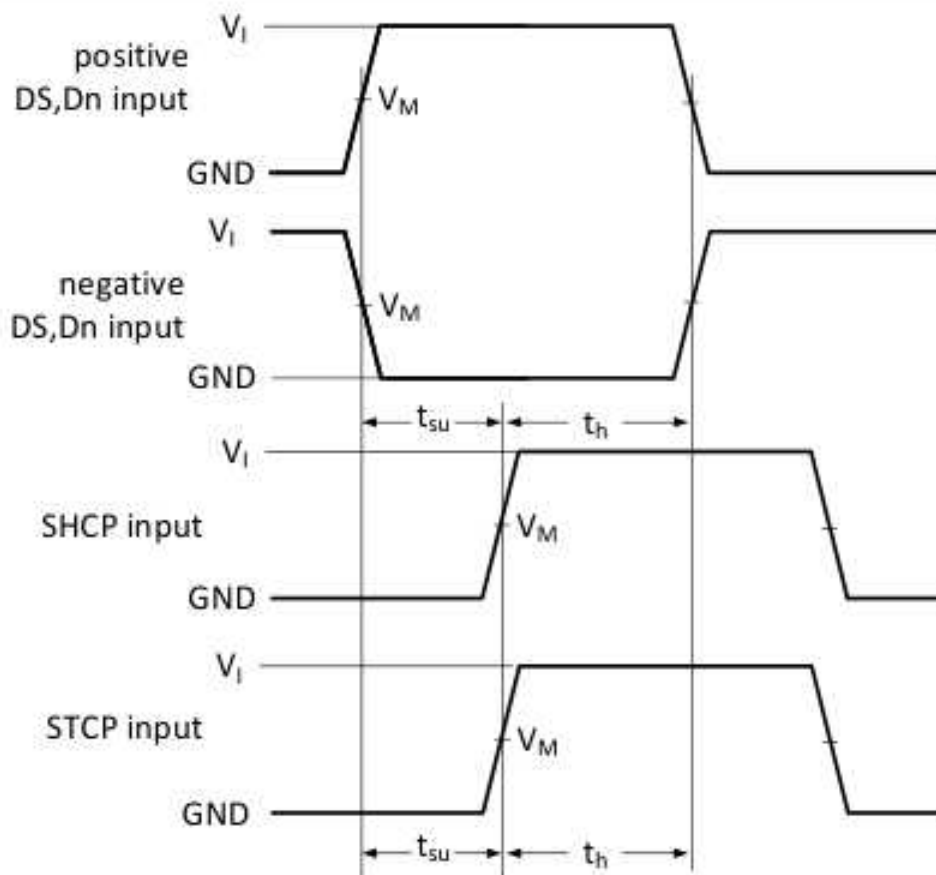
$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 9. Input  $\overline{\text{PL}}$  to Q output propagation delays,  $\overline{\text{PL}}$  pulse width and output transition times



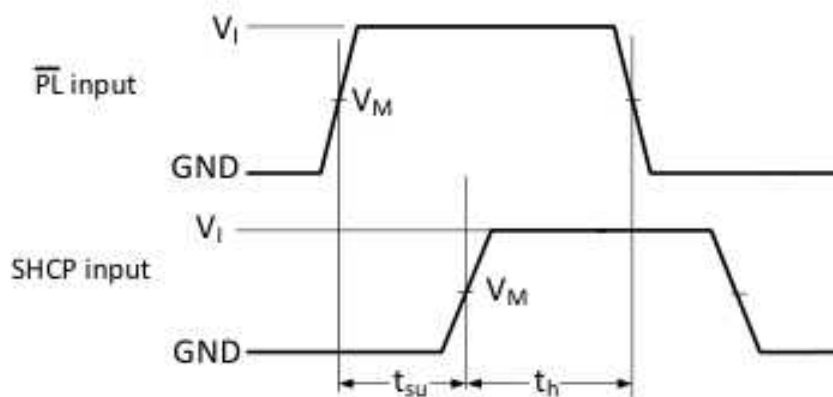
Measurement points are given in Table 8.

Fig. 10. Input  $\overline{\text{MR}}$  to shift clock SHCP and storage clock STCP recovery times



Measurement points are given in Table 8.

Fig. 11. Set-up and hold times for DS, Dn inputs to SHCP, STCP inputs



Measurement points are given in Table 8.

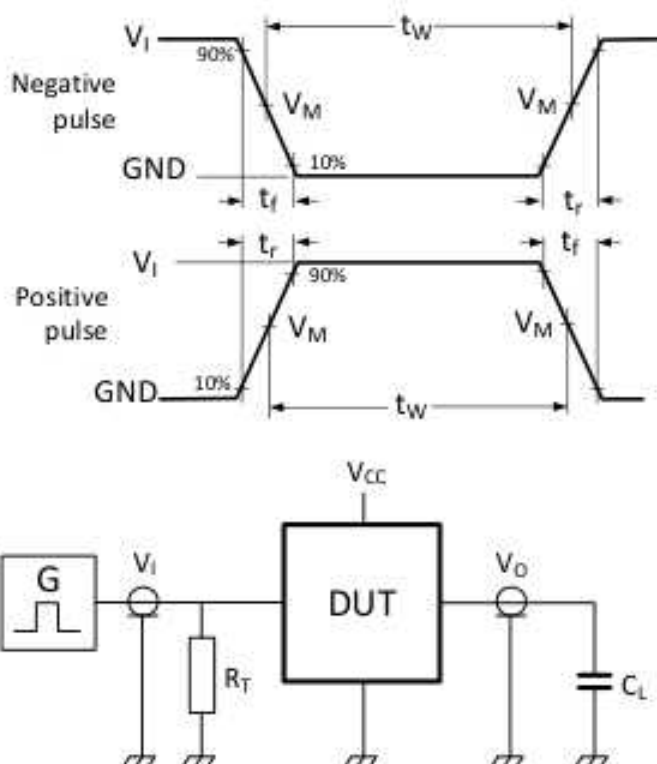
Fig. 12. Set-up and hold times for  $\overline{PL}$  input to SHCP input

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8-bit shift register with input flip-flops

**Table 8. Measurement points**

| Type       | Input              |                        | Output             |
|------------|--------------------|------------------------|--------------------|
|            | $V_M$              | $V_I$                  | $V_M$              |
| EM74HC597  | 0.5V <sub>CC</sub> | GND to V <sub>CC</sub> | 0.5V <sub>CC</sub> |
| EM74HCT597 | 1.3 V              | GND to 3 V             | 1.3 V              |



Test data is given in Table 9.

Definitions for test circuit:

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

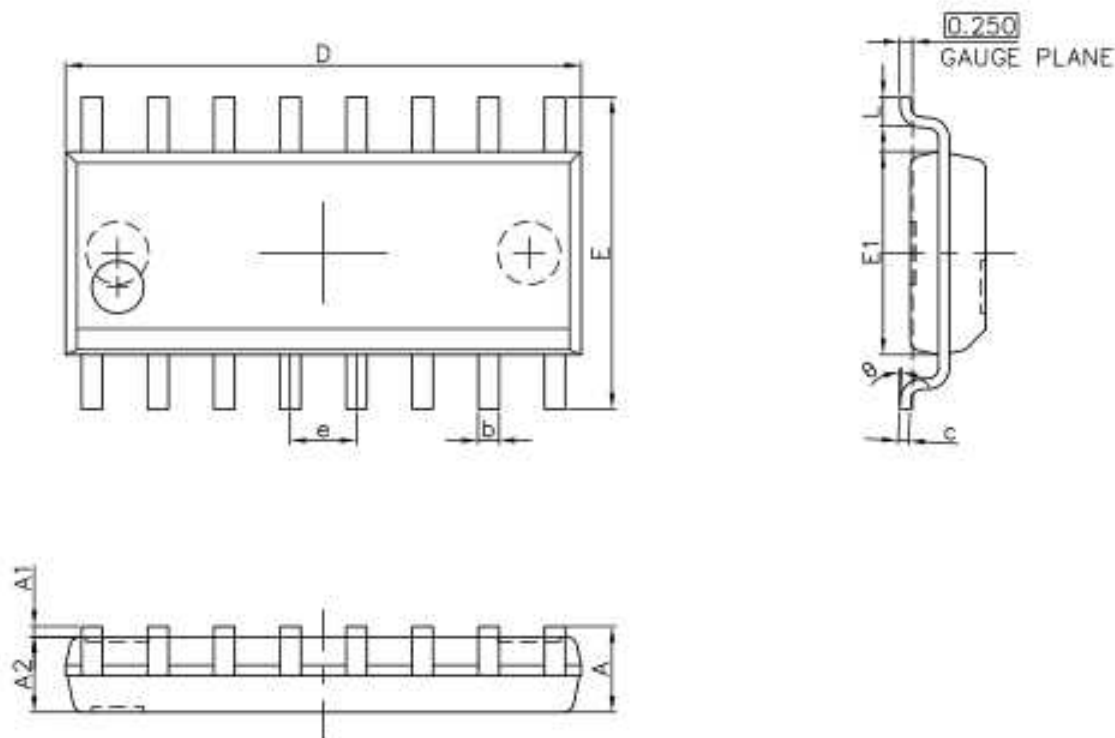
**Fig. 13. Test circuit for measuring switching times**

**Table 9. Test data**

| Type       | Input           |             | Load  | Test               |
|------------|-----------------|-------------|-------|--------------------|
|            | $V_I$           | $t_r = t_f$ | $C_L$ |                    |
| EM74HC597  | V <sub>CC</sub> | 2.5 ns      | 50 pF | $t_{PHL}, t_{PLH}$ |
| EM74HCT597 | 3 V             | 2.5 ns      | 50 pF | $t_{PHL}, t_{PLH}$ |

## 12. Package Outline

SOP-16L

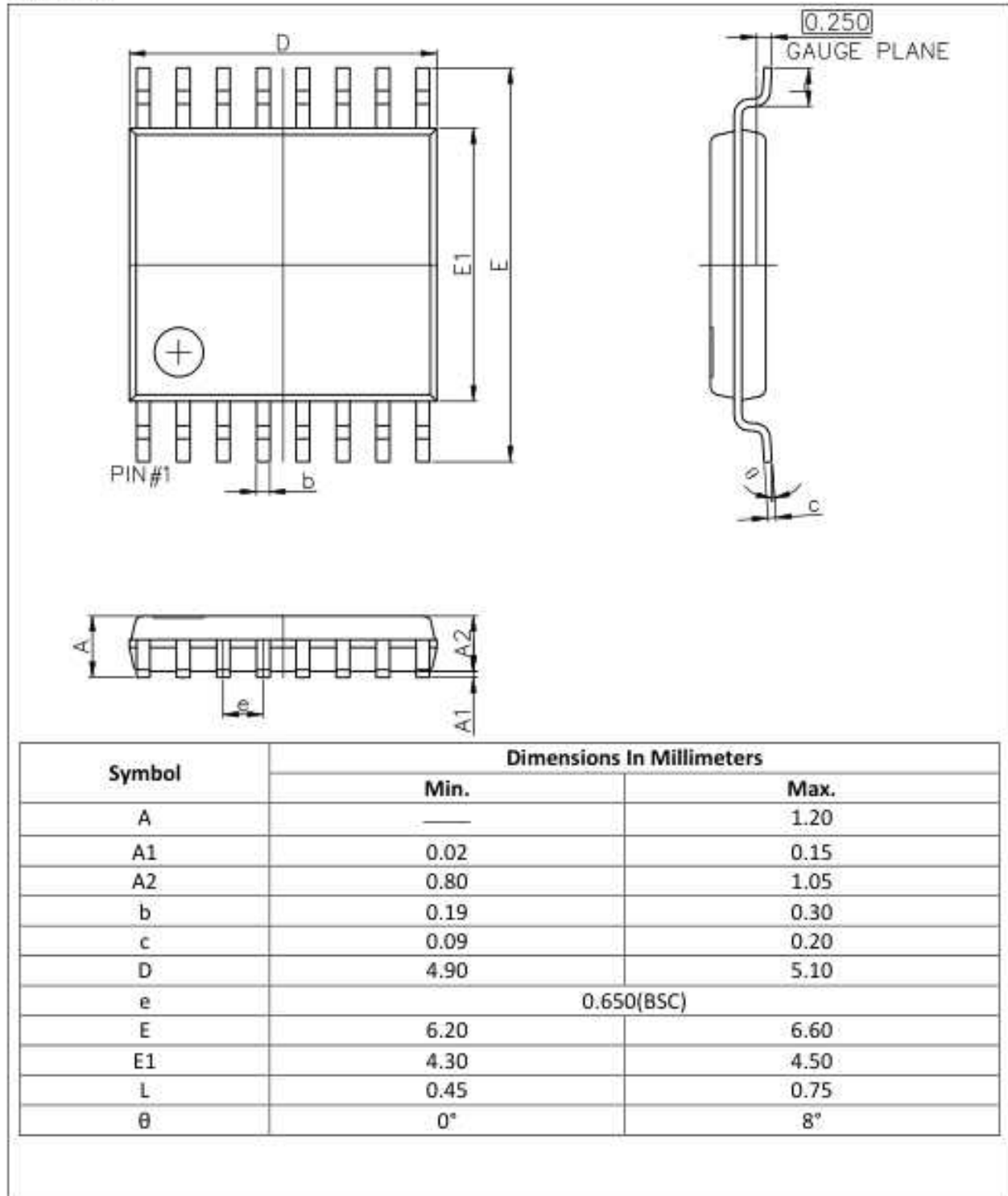


| Symbol | Dimensions In Millimeters |       |
|--------|---------------------------|-------|
|        | Min.                      | Max.  |
| A      | —                         | 1.75  |
| A1     | 0.05                      | 0.25  |
| A2     | 1.30                      | 1.60  |
| b      | 0.33                      | 0.51  |
| c      | 0.17                      | 0.25  |
| D      | 9.80                      | 10.00 |
| e      | 1.27(BSC)                 |       |
| E      | 5.80                      | 6.20  |
| E1     | 3.80                      | 4.00  |
| L      | 0.40                      | 1.27  |
| θ      | 0°                        | 8°    |

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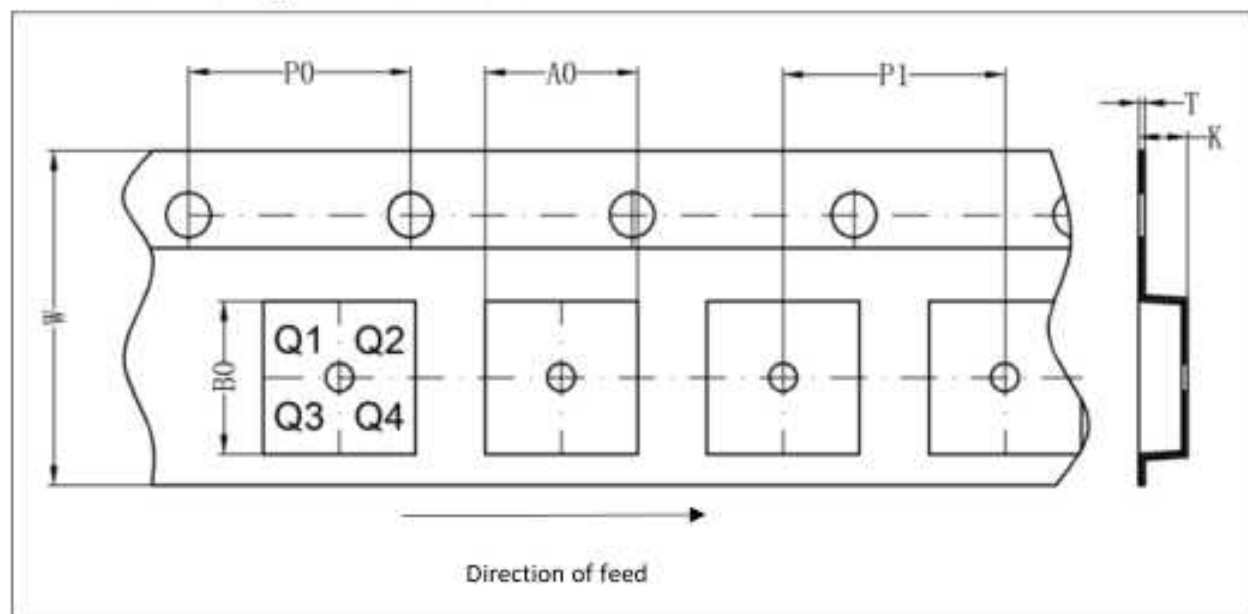
8-bit shift register with input flip-flops

## TSSOP-16L



## 13. Tape and Reel Information

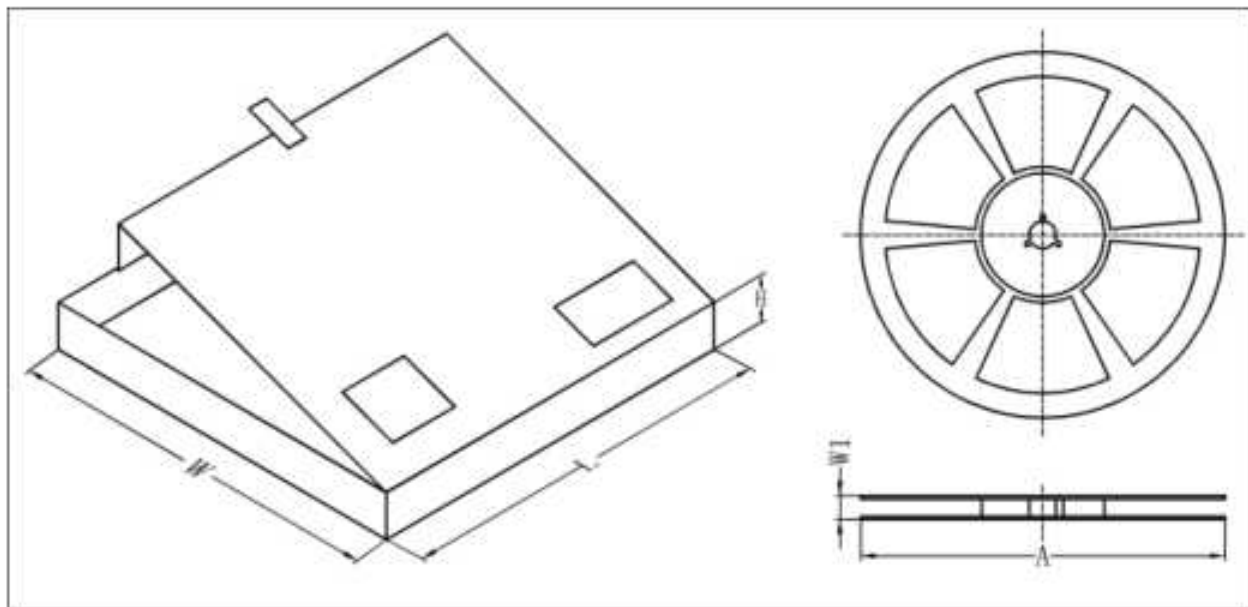
### 13.1. Carrier tape dimensions



**Table 10. Carrier tape dimensions**

| Package version | A0(mm) | B0(mm) | K0(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|--------|-------|--------|-------|--------|-------|
| SOP-16L         | 6.5    | 10.45  | 2.1    | 0.22  | 8      | 16    | 4      | Q1    |
| TSSOP-16L       | 6.7    | 5.45   | 1.6    | 0.25  | 8      | 12    | 4      | Q1    |

## 13.2. Reel and box dimensions



**Table 11. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | SPQ (pcs) [1] | Reels per box | Outer box dimensions L×W×H(mm) [2] |
|-----------------|-----------------|-----------------------|--------------------|---------------|---------------|------------------------------------|
| SOP-16L         | D               | 330                   | 22.4               | 3000          | 1             | 358x340x50                         |
| TSSOP-16L       | PW              | 330                   | 18.4               | 3000          | 1             | 358x340x50                         |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 14. Abbreviations

**Table 12. Abbreviations**

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| TTL     | Transistor-Transistor Logic             |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |

## 15. Revision History

**Table 13. Revision history**

| Document ID            | Release Date                                                                                                                                 | Data sheet status | Change notice | Supersedes             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|------------------------|
| EM74HC_HCT597 Rev. 1.1 | Mar 20, 2026                                                                                                                                 | Product datasheet |               | EM74HC_HCT597 Rev. 1.0 |
| Modifications:         | <ul style="list-style-type: none"> <li>Section 12: Package outline updated.</li> <li>Section 13: Added tape and reel information.</li> </ul> |                   |               |                        |
| EM74HC_HCT597 Rev. 1.0 | Nov 29, 2024                                                                                                                                 | Product datasheet |               |                        |